BBTV-Data_Exploration

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2025-09-18

Load libraries

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.5
## v forcats   1.0.0     v stringr   1.5.1
## v ggplot2   4.0.0     v tibble    3.2.1
## v lubridate 1.9.4     v tidyr     1.3.1
## v purrr     1.1.0
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
##
## Attaching package: 'stringdist'
##
##
## The following object is masked from 'package:tidyr':
##
##     extract
##
##
## Linking to GEOS 3.13.0, GDAL 3.8.5, PROJ 9.5.1; sf_use_s2() is TRUE
##
##
## Attaching package: 'janitor'
##
##
## The following objects are masked from 'package:stats':
##
##     chisq.test, fisher.test
##
##
##
## Attaching package: 'rnaturalearthdata'
##
##
## The following object is masked from 'package:rnaturalearth':
##
##     countries110
##
##
## -- Attaching packages -------------------------------------- tidymodels 1.4.1 --
##
```

```
## v broom        1.0.10     v rsample      1.3.1
## v dials        1.4.2      v tailor       0.1.0
## v infer        1.0.9      v tune         2.0.0
## v modeldata    1.5.1      v workflows    1.3.0
## v parsnip      1.3.3      v workflowsets 1.1.1
## v recipes      1.3.1      v yardstick    1.3.2
##
## -- Conflicts ------------------------------------------ tidymodels_conflicts() --
## x scales::discard()     masks purrr::discard()
## x stringdist::extract() masks tidyr::extract()
## x dplyr::filter()       masks stats::filter()
## x recipes::fixed()      masks stringr::fixed()
## x dplyr::lag()          masks stats::lag()
## x yardstick::spec()     masks readr::spec()
## x recipes::step()       masks stats::step()
##
##
## Attaching package: 'vip'
##
##
## The following object is masked from 'package:utils':
##
##     vi
##
##
##
## Attaching package: 'xgboost'
##
##
## The following object is masked from 'package:dplyr':
##
##     slice
##
##
## Loading required package: viridisLite
##
##
## Attaching package: 'viridis'
##
##
## The following object is masked from 'package:scales':
##
##     viridis_pal
##
##
## Loading required package: Rcpp
##
##
## Attaching package: 'Rcpp'
##
##
## The following object is masked from 'package:rsample':
##
##     populate
```

```
##
##
## Loading 'brms' package (version 2.22.0). Useful instructions
## can be found by typing help('brms'). A more detailed introduction
## to the package is available through vignette('brms_overview').
##
##
## Attaching package: 'brms'
##
##
## The following object is masked from 'package:dials':
##
##     mixture
##
##
## The following object is masked from 'package:stats':
##
##     ar
##
##
## This is posterior version 1.6.0
##
##
## Attaching package: 'posterior'
##
##
## The following objects are masked from 'package:stats':
##
##     mad, sd, var
##
##
## The following objects are masked from 'package:base':
##
##     %in%, match
```

Load data

Data cleaning

Note:

There are several qualitive variable that are transformed into a seemingly quantitative such as * Severity

1 = no symptoms
6 = bunchy top appearance
4 = chlorosis of leaf margin
2 = dark green streaks on leaf lamina 5 = reduction in leaf size
3 = dark green streaks on the petiole

- Disease_state

Asymptomatic = 0
Symptomatic = 1

Sanitation of the disease status

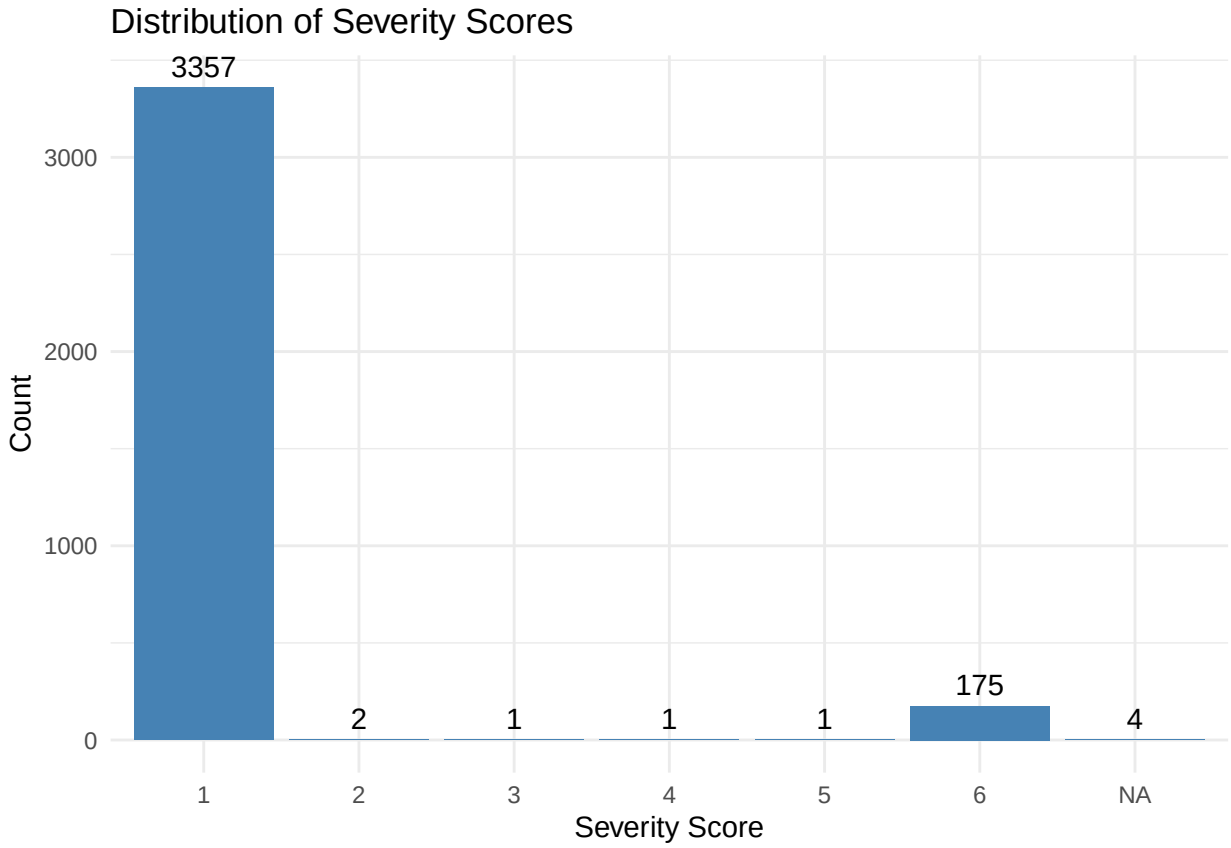
Summary statistic

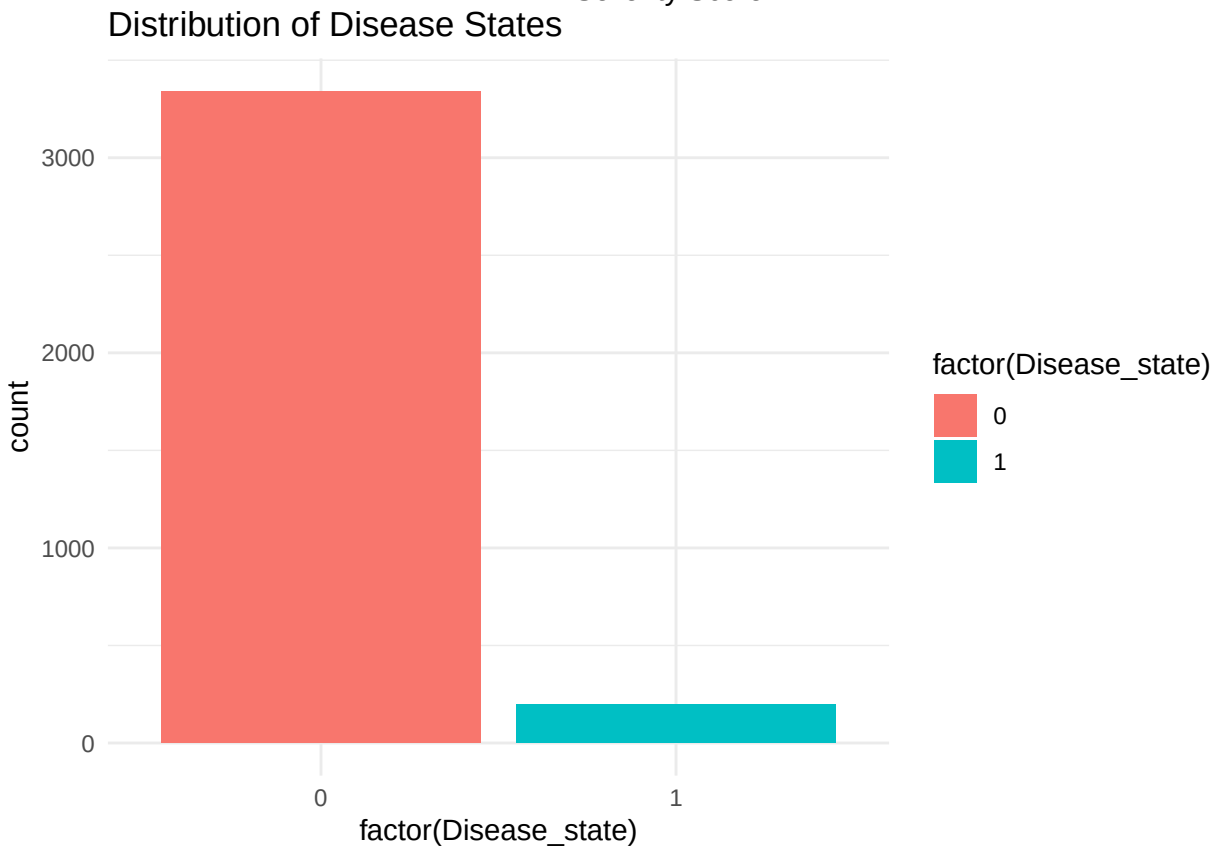
```
## -- Data Summary ------------------------
##                           Values
## Name                      Piped data
## Number of rows            3541
## Number of columns         16
## _______________________
## Column type frequency:
##   numeric                 16
## ________________________
## Group variables           None
##
## -- Variable type: numeric -------------------------------------------------------
##    skim_variable      n_missing complete_rate       mean        sd    p0    p25
##  1 TotalNsucker              27         0.992    6.08       5.13    0      2
##  2 SurveyedS                 27         0.992    2.73       3.35    0      1
##  3 AphidOwEL                  6         0.998    0.0141     0.353   0      0
##  4 AphidEL                   14         0.996    0.00680    0.187   0      0
##  5 AphidOwN                   4         0.999    0.00226    0.0672  0      0
##  6 AphidN                   180         0.949    0.00446    0.153   0      0
##  7 AphidOwB                  17         0.995    0.0610     1.72    0      0
##  8 AphidB                    15         0.996    0.0635     1.73    0      0
##  9 Fieldsize                 80         0.977 2180.      6074.     20    300
## 10 PlantationAge             60         0.983  120.        93.8     5     48
## 11 Distance                 139         0.961   10.6        9.10    0      4
## 12 NumberCountedField        60         0.983    7.37       8.99    0      2
## 13 Latitude                  60         0.983    7.45       1.23    6.25   6.63
## 14 Longitude                 60         0.983    2.27       0.355   1.56   1.96
## 15 Altitude                  60         0.983  157.       116.     -6.5   62.5
## 16 CoordPrecision            60         0.983   16.7       19.2     3      4.83
##       p50     p75     p100 hist
##  1   5       8       54
##  2   2       3       54
##  3   0       0       12
##  4   0       0        8
##  5   0       0        3
##  6   0       0        8
##  7   0       0      100
##  8   0       0      100
##  9 500    1000    50000
## 10 120     180      360
## 11   8      13       55
## 12   5       9       78
## 13   6.93    7.90    11.1
## 14   2.29    2.60     3.16
## 15 105.    221.     460.
## 16   8      22.8    148
```

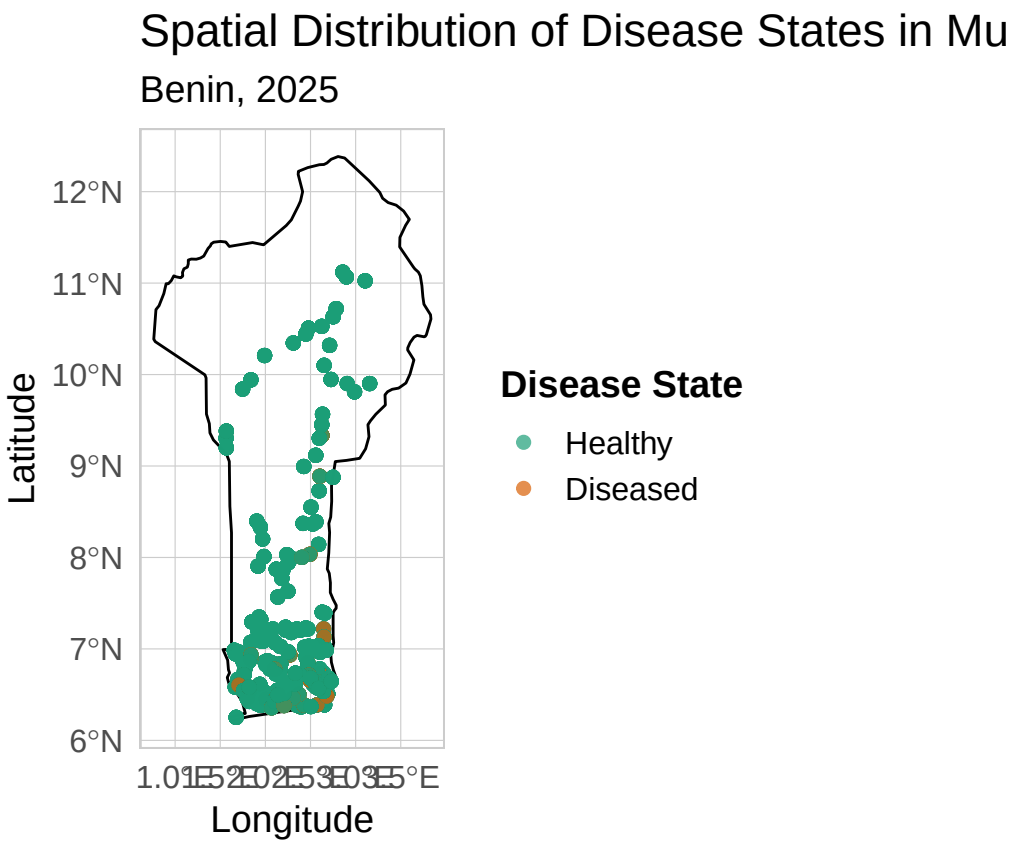
```
## -- Data Summary ------------------------
##                           Values
## Name                      Piped data
```

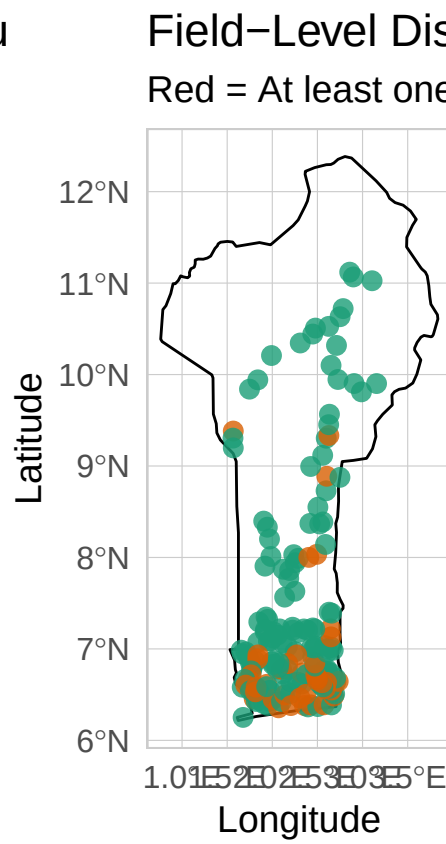
```
## Number of rows             3541
## Number of columns          40
## _______________________
## Column type frequency:
##   character                7
##   factor                   33
## ________________________
## Group variables            None
##
## -- Variable type: character ----------------------------------------------------
##   skim_variable   n_missing complete_rate min max empty n_unique whitespace
## 1 Unique_Field_ID         0       1         4   8     0      176          0
## 2 PlantNumber             0       1         3   3     0       30          0
## 3 Symptoms               92       0.974     8  71     0        5          0
## 4 Departement            60       0.983     3  11     0       14          0
## 5 Commune               372       0.895     2  14     0       56          0
## 6 Location               80       0.977     2  19     0      168          0
## 7 TypeFarming          3511       0.00847  38  38     0        1          0
##
## -- Variable type: factor -------------------------------------------------------
##    skim_variable     n_missing complete_rate ordered n_unique
##  1 Disease_state             0         1     FALSE          2
##  2 Severity                  4         0.999 FALSE          6
##  3 Variety                   5         0.999 FALSE          3
##  4 CMV                       4         0.999 FALSE          2
##  5 BSV                       3         0.999 FALSE          2
##  6 BXW                       4         0.999 FALSE          2
##  7 Sigatoka                  7         0.998 FALSE          2
##  8 Fusarium Wilt             5         0.999 FALSE          2
##  9 AgroEco                  60         0.983 FALSE          4
## 10 Intercrops               60         0.983 FALSE          2
## 11 Management               60         0.983 FALSE          2
## 12 Weediness                60         0.983 FALSE          2
## 13 CS_Papayer                0         1     FALSE          2
## 14 CS_Palmier                0         1     FALSE          2
## 15 CS_manguier               0         1     FALSE          2
## 16 CS_Amarante               0         1     FALSE          2
## 17 CS_Eucalyptus             0         1     FALSE          2
## 18 CS_cocotiers              0         1     FALSE          2
## 19 CS_Citronnelle            0         1     FALSE          2
## 20 CS_Ananas                 0         1     FALSE          2
## 21 CS_Manioc                 0         1     FALSE          2
## 22 CS_Vernonia               0         1     FALSE          2
## 23 CS_basilicum              0         1     FALSE          2
## 24 CS_Maize                  0         1     FALSE          2
## 25 CS_Canne à sucre          0         1     FALSE          2
## 26 CS_Basilic                0         1     FALSE          2
## 27 CS_vernonia               0         1     FALSE          2
## 28 CS_grande morelle         0         1     FALSE          2
## 29 CS_Assissrè               0         1     FALSE          2
## 30 CS_Igname                 0         1     FALSE          2
## 31 CS_Gombo                  0         1     FALSE          2
## 32 CS_arachide               0         1     FALSE          2
## 33 CS_haricot                0         1     FALSE          2
```

```
##    top_counts
##  1 0: 3342, 1: 199
##  2 1: 3357, 6: 175, 2: 2, 3: 1
##  3 Swe: 2063, Ban: 1469, loc: 4
##  4 0: 3532, 1: 5
##  5 0: 3521, 1: 17
##  6 0: 3532, 1: 5
##  7 0: 3520, 1: 14
##  8 0: 3528, 1: 8
##  9 6: 1560, 4: 1201, 5: 360, 7: 360
## 10 0: 2755, 1: 726
## 11 1: 1989, 0: 1492
## 12 1: 2128, 0: 1353
## 13 0: 3491, 1: 50
## 14 0: 3451, 1: 90
## 15 0: 3511, 1: 30
## 16 0: 3501, 1: 40
## 17 0: 3511, 1: 30
## 18 0: 3511, 1: 30
## 19 0: 3521, 1: 20
## 20 0: 3521, 1: 20
## 21 0: 3444, 1: 97
## 22 0: 3521, 1: 20
## 23 0: 3521, 1: 20
## 24 0: 3262, 1: 279
## 25 0: 3481, 1: 60
## 26 0: 3521, 1: 20
## 27 0: 3521, 1: 20
## 28 0: 3521, 1: 20
## 29 0: 3521, 1: 20
## 30 0: 3501, 1: 40
## 31 0: 3531, 1: 10
## 32 0: 3521, 1: 20
## 33 0: 3521, 1: 20
```


Distribution of Severity Scores
3357
2
1
1
1
175
4
3000
2000
1000
0
Count
1
2
3
4
5
6
NA
Severity Score


Distribution of Disease States
3000
2000
1000
0
count
0
1
factor(Disease_state)
factor(Disease_state)
0
1





What can explain the occurence of the BBTD?

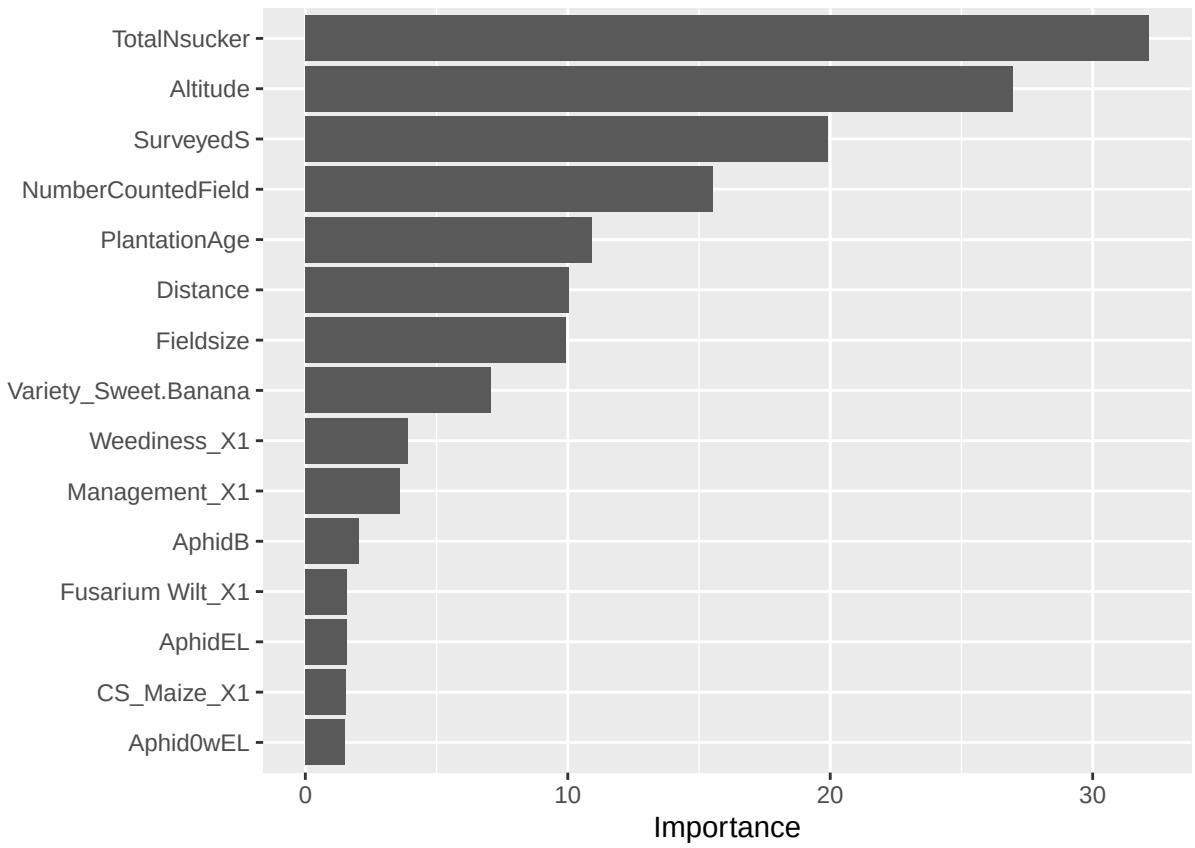
```
## i Creating pre-processing data to finalize 1 unknown parameter: "mtry"

## > A | warning: ! 44 columns were requested but there were 43 predictors in the data.
##                 i 43 predictors will be used.

## There were issues with some computations   A: x1There were issues with some computations   A: x1

## # A tibble: 3 x 4
##   .metric     .estimator .estimate .config
##   <chr>       <chr>          <dbl> <chr>
## 1 accuracy    binary        0.928  pre0_mod0_post0
## 2 roc_auc     binary        0.916  pre0_mod0_post0
## 3 brier_class binary        0.0499 pre0_mod0_post0

##            Truth
## Prediction   0   1
##          0 573  36
##          1   9   9
```



With another framework more powerfull. Let us use the xgboost

What is XGBoost?

XGBoost (Extreme Gradient Boosting) is a powerful machine learning algorithm that uses boosted decision trees.

- Compared to Random Forest:

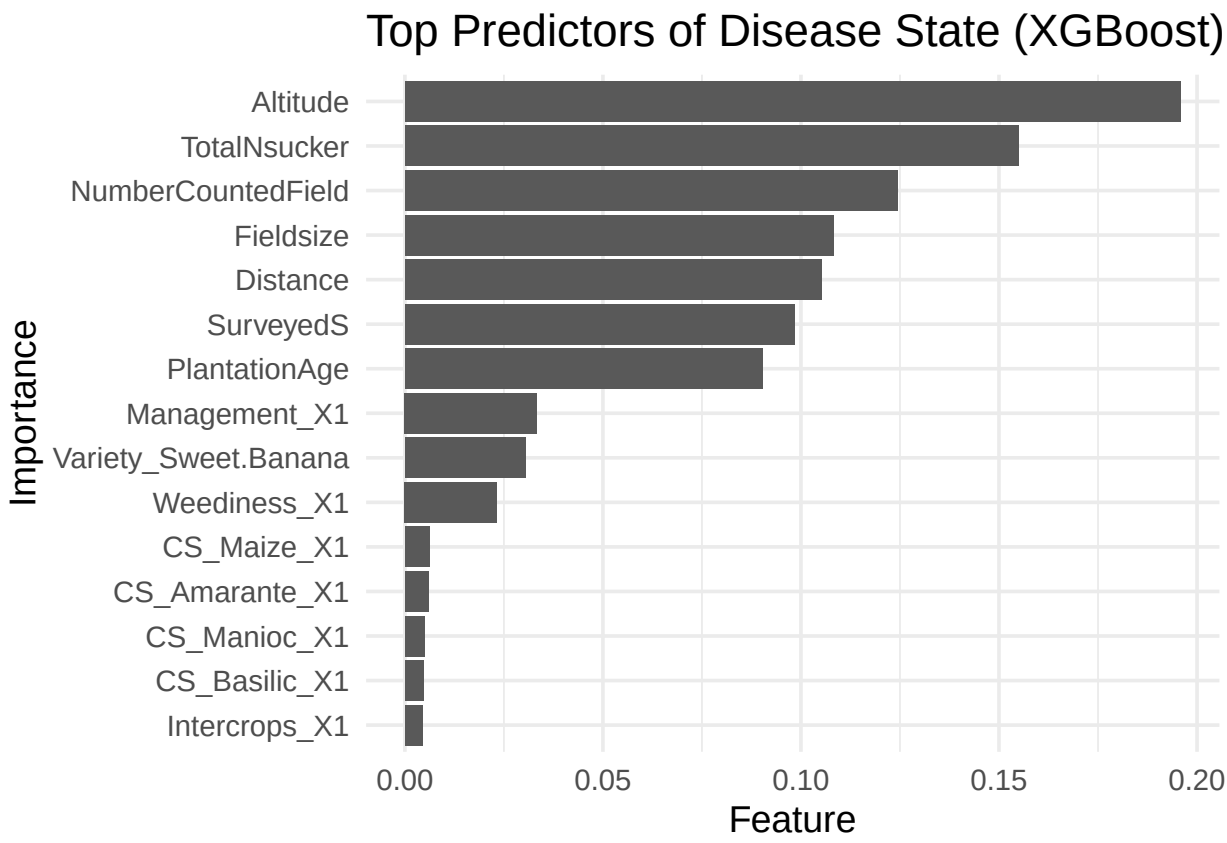
RF builds many independent trees (bagging).

XGBoost builds trees sequentially, where each new tree corrects errors of the previous ones (boosting).

It's fast, handles missing values, works well with tabular data, and often outperforms RF in classification problems like your disease-state prediction.

```
## i Creating pre-processing data to finalize 1 unknown parameter: "mtry"
```

```
## # A tibble: 3 x 4
##   .metric     .estimator .estimate .config
##   <chr>       <chr>          <dbl> <chr>
## 1 accuracy    binary        0.928  pre0_mod0_post0
## 2 roc_auc     binary        0.919  pre0_mod0_post0
## 3 brier_class binary        0.0502 pre0_mod0_post0
```

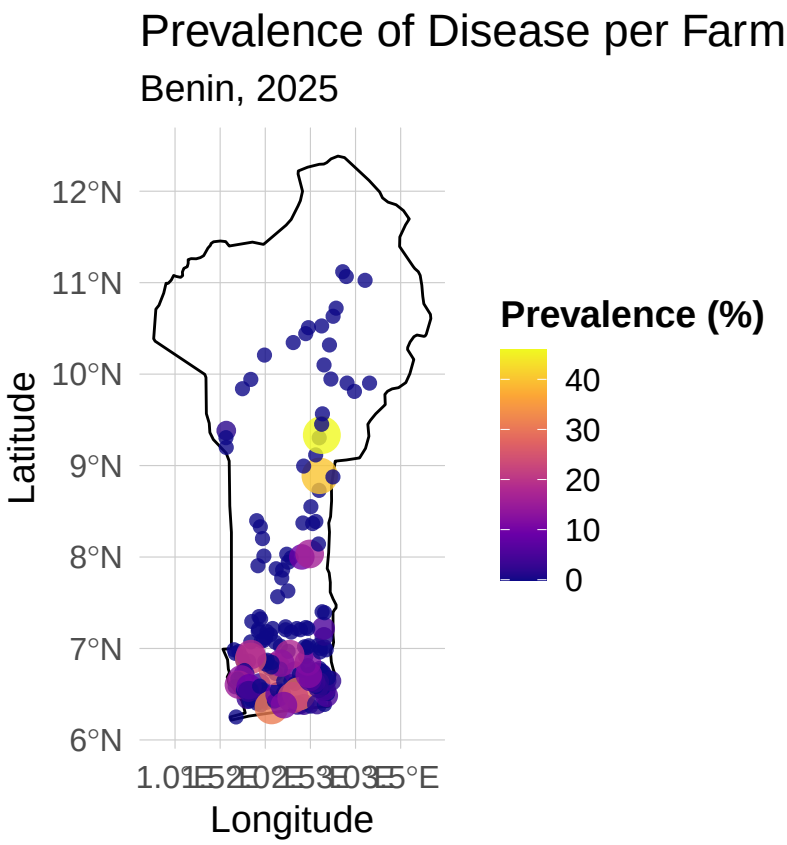


Compute the prevalence per farm

```
## # A tibble: 176 x 4
##    Unique_Field_ID diseased_plants total_surveyed prevalence
##    <chr>                     <int>          <dbl>      <dbl>
##  1 F0147_0                      11             24       45.8
##  2 F0142_0                      12             30       40
##  3 F0111_0                       9             29       31.0
##  4 F0124_0                       7             24       29.2
##  5 F0118_0                       7             27       25.9
##  6 F087_0                        9             38       23.7
##  7 F0125_0                       6             26       23.1
##  8 F088_0                        7             39       17.9
##  9 F071_0                       12             67       17.9
## 10 F0135_1                      13             80       16.2
## # i 166 more rows
```

```
## `summarise()` has grouped output by 'Unique_Field_ID', 'Latitude'. You can
## override using the `.groups` argument.
```

Plot on a map



Detecting the epidemic front

- Concept: The epidemic front is the leading edge where disease prevalence begins to appear or sharply increases across the landscape.

Detection of the true epidemic size

Concept & model (quick description + formulas)

We observe, for each farm i. let us denote y_i number of diseased plants observed (aggregated per farm); n_i, number of plants surveyed (SurveyedS) at that farm; X_i farm covariates that explain true prevalence (e.g., Variety, Fieldsize, PlantationAge, altitude, vector counts…) and $W - i$ the effort / detection covariates (e.g., Distance travelled per farm, # farms visited per km proxy, time spent per farm, SurveyedS itself, collector id if available).

We assume two latent quantities:

- p_i the true prevalence (probability a plant is diseased) at farm i. Modelled with a logistic regression:

$$logit(p_i) = X_i\beta + \mu_i$$

where μ_i is the random effect or a spatial/random field (optional). The detection probability d_i i.e. probability a diseased plant is recorded given it exists) at farm i.

$$logit(d_i) = W_j\gamma$$

The observed counts arise because only a fraction of diseased plants are detected:

$$y_i \sim Binomial(n_i, p_i \times d_i)$$

The **Epidemic size** (total infected plants in study region): if you know (or can estimate) N_i number of plants on farm i then the total number infected (T) is:

$$T = \sum_i N_i p_i$$

Using Bayesian posterior draws of p_i we compute posterior draws of T to quantify uncertainty (credible intervals).

If we only want the total expected proportion infected across surveyed farms:

$$\hat{p} = \frac{\sum_i N_i p_i}{\sum_i n_i}$$

```
## # A tibble: 1 x 10
##   Unique_Field_ID Latitude Longitude Fieldsize PlantationAge Variety    Distance
##   <chr>              <dbl>     <dbl>     <dbl>         <dbl> <fct>         <dbl>
## 1 F093_0              6.58      1.67      2000           120 Banana pl~       10
## # i 3 more variables: diseased <int>, surveyed <dbl>, farms_per_km <dbl>
```

```
## Compiling Stan program...
```

```
## Trying to compile a simple C file
```

```
## Running /Library/Frameworks/R.framework/Resources/bin/R CMD SHLIB foo.c
## using C compiler: ‘Apple clang version 17.0.0 (clang-1700.0.13.5)’
## using SDK: ‘MacOSX15.5.sdk’
## clang -arch arm64 -I"/Library/Frameworks/R.framework/Resources/include" -DNDEBUG   -I"/Library/Framew
## In file included from <built-in>:1:
## In file included from /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/library/StanHeade
## In file included from /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/library/RcppEigen/
## In file included from /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/library/RcppEigen/
## /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/library/RcppEigen/include/Eigen/src/Core
##   679 | #include <cmath>
##       |          ^~~~~~
## 1 error generated.
## make: *** [foo.o] Error 1
```

```
## Start sampling
```

```
## [1] 232
```

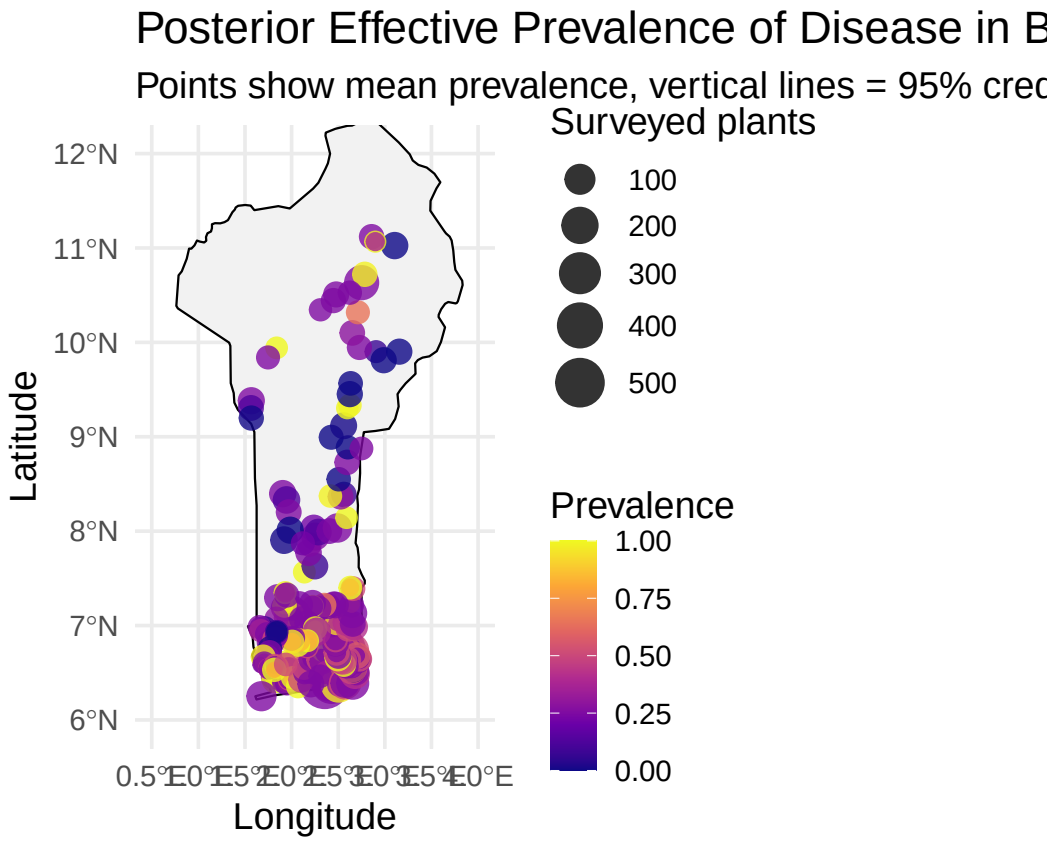
```
## [1] 4000  232
```

```
## [1] 5
```

```
##  [1]   2   3   4   5 111  NA  NA  NA  NA  NA  NA  NA  NA  NA  NA  NA  NA  NA  NA
## [20]  NA
```

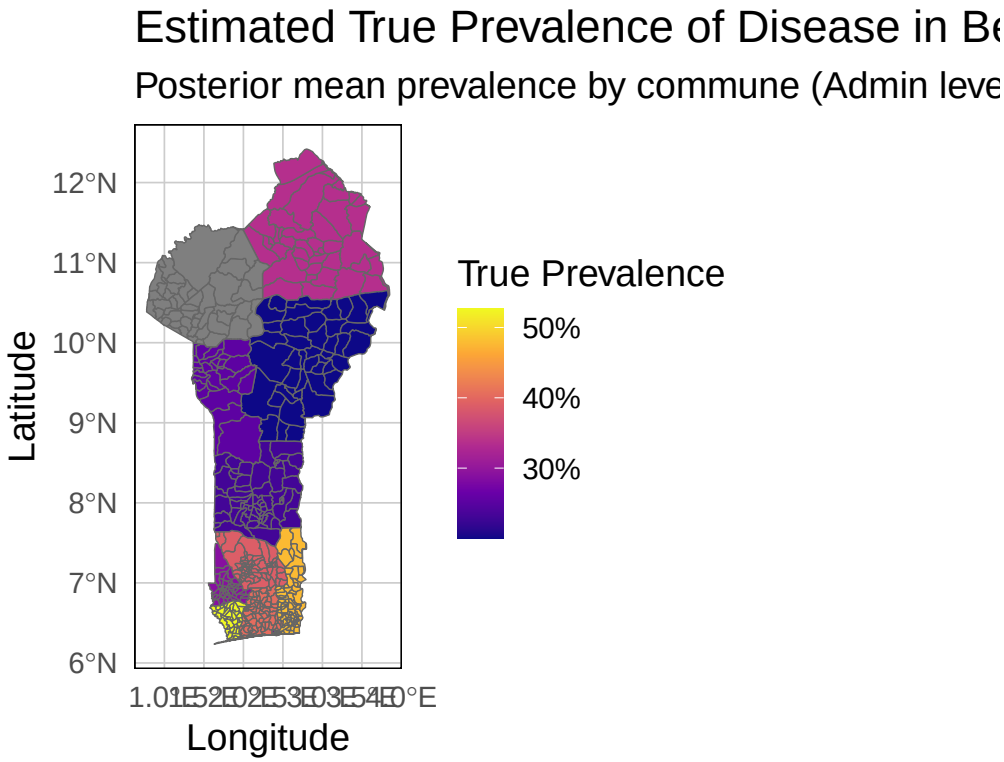
Viz

Viz prevalance instead



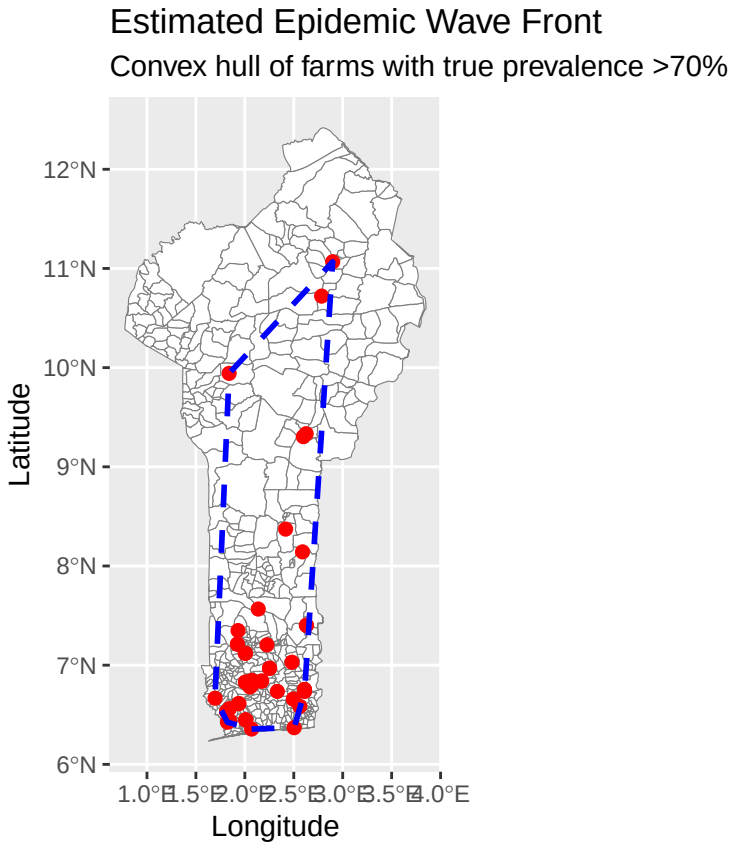
At the level 3

```
## Reading layer `gadm41_BEN_3' from data source
##   `/Users/koissi/Desktop/WAVE_project/Theory/CassavaMap/benin_shp/gadm41_BEN_3.shp'
##   using driver `ESRI Shapefile'
## Simple feature collection with 546 features and 16 fields
## Geometry type: MULTIPOLYGON
## Dimension:     XY
## Bounding box:  xmin: 0.774345 ymin: 6.23491 xmax: 3.851701 ymax: 12.41835
## Geodetic CRS:  WGS 84
```



show estimated prevalence adjusted for detection.
95% CI available in supplementary table.

Detection of the wave front of the epidemcis



Method in R:

Use spatial interpolation (e.g., kriging or inverse distance weighting) to map prevalence continuously across the landscape.

Identify the contour line corresponding to a low prevalence threshold (e.g., 5–10%) as the "front."

Is there any spatial cluster within the data?